21. Current Dataset: Collection and Coverage

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

5 people **25 kept recordings** **2,050,030 raw rows**

Recording block	Files	CSI rows
Actions: main block	5	844,426
Actions: seated	2	173,297
Signs: standing	5	457,408
Signs: seated	2	127,209
Interleaved stories	5	354,465
Still-person baselines	3	54,094
Empty-room checks	3	39,131
Total collected	**25**	**2,050,030**

Model input: **22 files / 2,010,899 rows**. Recording totals include rest/ready; empty-room files are QC only.

Room, transmitter and receiver

collection_setup.png

Collection conditions

Two USRP N210s; Channel 1, 2412 MHz.

Same room and antenna positions.

One active participant at a time.

P4/P5 add seated blocks and cooking actions.

22. Current Dataset: Actions and Signs

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Isolated action: pick up an object

`collection_action_pickup.png`

Seated sign: COOK

`collection_sign_cook_seated.png`

12 action labels

Walk	Sit–stand
Pick up	Light switch
Drink	Answer phone
Eat	Wipe table
TV remote	Fall
Chop food	Stir pot

628 isolated trials. P4/P5 add chop/stir and seven seated action types.

14 sign labels

HELLO	THANKYOU
LAMP	FAN
TELEVISION	COOK
WATER	HOT
HUNGRY	TOILET
CALL	HELP
PAIN	ACCIDENT

490 trials: 350 standing (P1–P5) + 140 seated (P4/P5).

23. Current Dataset: Interleaved Stories

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.

Template	Recorded sequence / context
M-SEQ: meal	Walk, light switch, sit–stand, eat/drink; HELLO, HUNGRY, COOK, HOT, WATER, THANKYOU. P4/P5 add chop/stir.
T-SEQ: TV	Walk, sit–stand, TV remote, phone; HELLO, TELEVISION, HOT, LAMP, CALL, TOILET; stand and walk out.
C-SEQ: clean / fall	Walk, wipe, pick up; FAN, HELP; walk and fall; HELP, PAIN, ACCIDENT, CALL on the mattress.
S-SEQ: seated	Wipe, chop, stir, eat, drink; HELLO, HUNGRY, COOK, HOT, WATER, HELP, THANKYOU. P4/P5 only.

Each available template was repeated twice per person: **34 runs / 416 labelled steps** = 208 actions + 208 signs.

Meal story: sign after an action

`collection_story_meal.png`

Why stories matter

Actions and signs share one recording.

HOT during a meal and HOT while watching TV have different intended responses.

24. Current Dataset: CSI Rows and Annotations

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Raw CSI file: one row per decoded packet

P04_signs_seated_20261016_174235.csv | 107 columns: 3 metadata + 52 × (Re, Im)

Packet	Timestamp_us	Channel	SC0_Re	SC0_Im	SC1_Re	SC1_Im	...
11676	1792152908290345	1	0.000275	-0.000327	0.001651	-0.000113	...
11677	1792152908312995	1	0.000357	-0.000239	0.001407	0.000079	...
11678	1792152908318294	1	0.001165	0.000000	0.001127	-0.000149	...
11679	1792152908386570	1	-0.001413	-0.000463	-0.000872	-0.001368	...

Separate label file: join each packet to its annotated time interval

pid	type	label	posture	t_start_us	t_end_us
P04	sign	ACCIDENT	seated	1792152899283128	1792152903283699
P04	sign	COOK	seated	1792152908284587	1792152912286562
P04	sign	TOILET	seated	1792152917287496	1792152921288077

Selected columns shown. Values are copied from the actual recording and label CSVs; other subcarrier columns are omitted.

Actual CSV excerpt, selected columns. Raw rows contain packet/time/channel and complex CSI; the separate label file supplies participant, type, label, posture and time boundaries. Labels come from the collection cues/key presses; video alignment is not yet verified.

25. Current Dataset: Recording Rows per Class

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

Actions

Class	Trials	Raw rows
Answer phone	67	46,741
Chop food	40	25,679
Drink	67	46,554
Eat	67	46,401
Fall	23	16,512
Light switch	48	23,817
Pick up	48	35,094
Sit–stand	48	33,576
Stir pot	40	25,174
TV remote	67	39,134
Walk	46	33,371
Wipe table	67	44,783
Total	**628**	**416,836**

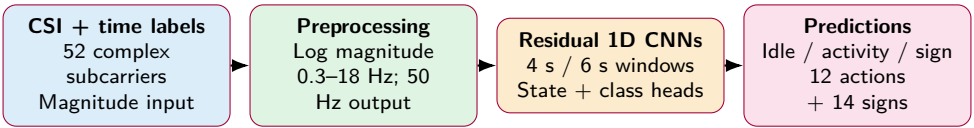
Signs

Class	Trials	Raw rows
ACCIDENT	35	15,864
CALL	35	15,815
COOK	35	15,792
FAN	35	15,956
HELLO	35	16,019
HELP	35	15,782
HOT	35	16,097
HUNGRY	35	16,140
LAMP	35	15,689
PAIN	35	16,051
TELEVISION	35	15,641
THANKYOU	35	15,697
TOILET	35	15,875
WATER	35	16,152
Total	**490**	**222,570**

Raw packets inside fully covered isolated label intervals; rest/ready and story steps excluded. Partial P2 trials excluded. Chop/stir: P4/P5 only.

31. Current Model: Training and Evaluation

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.



Five held-out-person tests

- Initial fit: 3 people; validation: 1 other person.
- Choose detector/action/sign roles on validation.
- Refit selected networks on 4 people, two seeds.
- Test only on the fifth person; repeat for all five.

All-five deployment model

Two normalized four-second CNNs, one per seed, trained on all five people.

Scales, masks and class order saved.

Kaggle T4 $\times 2$: **41.4 minutes**.

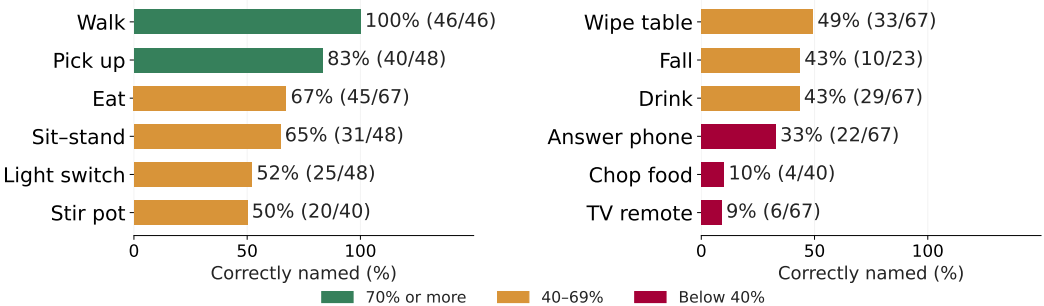
Scores come from held-out folds; the all-five refit has no independent score. Inference is offline. ExtraTrees applies only to isolated action fusion (final weight: zero).

32. Current Results: Action Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

311/628 correct (49.5% accuracy)

Mean held-out macro F1: **50.2%**



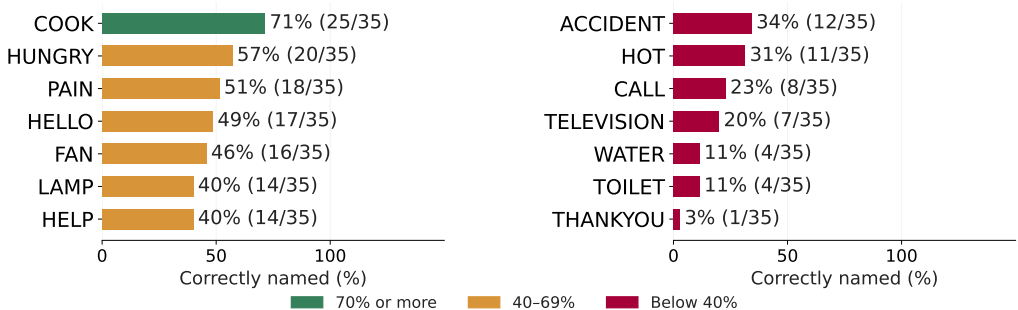
P1–P5 isolated trials, known boundaries/type; bars show per-class recall. **Strongest:** walk/pick up. **Weakest:** TV remote/chop/phone. Chop/stir cover P4/P5 only.

33. Current Results: Sign Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.

171/490 correct (34.9% accuracy)

Mean held-out macro F1: **33.2%**



P1–P5: 350 standing + 140 seated trials; 35 per sign; known boundaries/type. **Stronger:** COOK/HUNGRY/PAIN. **Weakest:** THANKYOU/WATER/TOILET. Isolated seated signs cover P4/P5 only.

34. Current Results: Detection Threshold

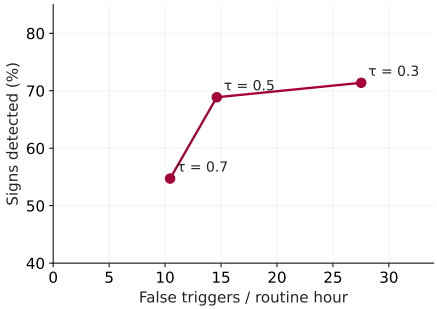
RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.

Setting	Signs caught	False / hour
$\tau = 0.3$: sensitive	71.4%	27.5
$\tau = 0.5$: middle	68.9%	14.6
$\tau = 0.7$: strict	54.7%	10.5

Raise τ to reduce false triggers; fewer signs are caught.

"Caught" means detected, not correctly named.

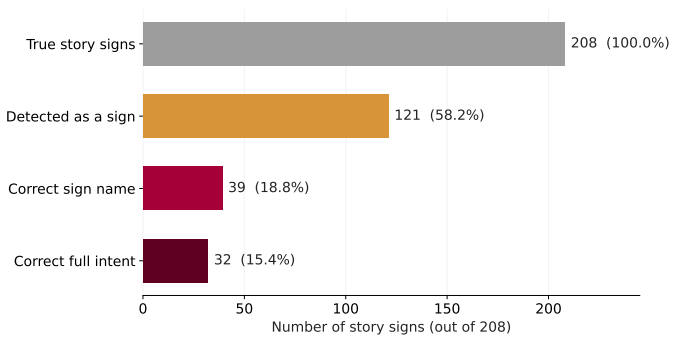
τ is a score cutoff, not calibrated certainty.



Fixed-threshold sweep: P1–P3, isolated + stories, 318 signs; mean over three people. One-second smoothing/minimum run, one-second gap merge and ± 1 s matching. Rates cover routine movement only (about 58 measured minutes), excluding idle/unknown periods.

35. Current Results: Continuous Story Recognition

RO1: Privacy-preserving single-modality sensing to separate daily Activities from Sign Gestures.
RO4: Multimodal fusion for context- and intent-aware smart-home control.



Automatic story inference

No true sign boundaries or type are supplied.

58.2% of signs detected.

18.8% correctly named.

15.4% correct full intent.

Continuous identity is the main remaining limitation.

All five people; validation-selected thresholds per fold; one-to-one IoU ≥ 0.10 matching. This differs from the fixed-threshold sweep. There were 50 false triggers during 20.3 minutes of measured story routine activity.